SBC 201 Chapter 21 Masonry — Group R-2 High-Rise Dimensional Requirements Matrix (2024)

1. Document metadata and use limitation

- **Project basis:** Riyadh, Saudi Arabia; Group R-2 residential high-rise; an occupied floor is stated to be more than 23 m above the relevant reference level.
- **Deliverable tier:** Project-use matrices in Sections 1–12 (design-check rows, not pasted inventory), plus a coverage summary and unresolved-source register. The full row inventory is not published.
- **Code/source basis:** SBC 201 (2024), Chapter 21, source file `Reference\SBC 201 2024\source_reference\Chapter_21 – MASONRY.txt`.
- **Extraction audit:** Skill extract. Project-use rows follow the chapter-extract row contract (noun-phrase checks, bold SI values, building-language triggers, named exceptions, check-specific actions). Internal inventory: **241** independently checkable numeric records (**202** Verified, **39** Verify source). Flattened appended tables are listed in the register and are not design-release values. Empirical adobe (2109) and dry-stack (2114) are inventoried for coverage only and are not opened in project-use tables.
- **Model:** Cursor Grok 4.6.
- **Prepared:** 2026-08-27.
- **Status:** Source-only architectural advisory matrix for design coordination. It is not a stamped compliance statement, structural design, SCD NOC, SBPS approval, legal opinion, or permission to construct.
- **Outbound-source rule:** No value in this matrix has been imported from SBC 305, SBC 301, SBC 303, SBC 304, TMS 504, TMS 604, TMS 602, Chapter 7 / Table 705.5, Chapter 14 veneer millimetres, Chapter 16 loads, Chapter 17 SI frequencies, Chapter 18 foundations, Section 2304.13, ANSI A108 / A118 manuals, ASTM product standards, commentary examples, or the existing chapter summary. Where Chapter 21 sends the user elsewhere, this matrix records the dependency without supplying the outbound value.

Scope and assumptions

1. Group R-2 and high-rise status are project statements, not independently verified classifications.
2. The exact Riyadh AHJ/permit pathway, project stage, fire-strategy status and SCD NOC status are unconfirmed; therefore this matrix does not conclude compliance.
3. Automatic sprinkler protection, mixed-use podium and storey count are unconfirmed. Chapter 21 does not branch on NFPA 13 versus 13R.
4. Seismic Design Category is unconfirmed. Fireplace/chimney seismic reinforcement in 2111.4 / 2113.3 is charged only for SDC **C or D**; SDC **A or B** does not require that reinforcement.
5. Structural masonry (load-bearing walls, shear walls, masonry beams) versus veneer-only versus nonstructural CMU partitions is unconfirmed. Member sizes, (f'_m), grout space, and most reinforcement rules live in **SBC 305**, not in this chapter.
6. Masonry veneer is charged to **Chapter 14** (2101.2.1). Do not take anchored/adhered veneer millimetres from this extract.
7. Empirical adobe (2109) is Type V, one- or two-storey construction and is not opened for this high-rise. Dry-stack masonry (2114) is not opened as a typical tower system.
8. All five appended tables are concatenated OCR. No reconstructed table cell is adopted as a design-release value.

2. Legends

Applicability

Status	Meaning
Direct	Expected to govern the stated R-2 tower basis, subject to confirmed geometry and design data.
Conditional	Governs only when the stated feature, load, sprinkler branch or exception exists.
Not typical	Unrelated occupancy-only rule; omitted from this deliverable unless the gap register already opened that use.
External verification	Chapter 21 points to another section/code/standard, or the project/AHJ basis must be confirmed before use.

Source confidence

Status	Meaning
Verified	Requirement and any stated numeric value were checked against unambiguous mandatory Chapter 21 source text.
Verify source	OCR, flattened table, page-split, or footnote attachment is unresolved. Not a design-release value.

3. Project decision and gap register

Decision / gap	Current project basis	Why it controls Chapter 21 application	Required project action
Structural engineer / SBC 305	Unconfirmed companion design	2101.2 charges masonry to this chapter **and SBC 305**; this extract has no wall thickness, (f'_m), grout space or standard lap tables	Issue an SBC 305 structural masonry package; do not size masonry from Chapter 21 alone
Masonry role on the tower	Unconfirmed: structural walls vs CMU partitions vs veneer-only	Veneer goes to Chapter 14; 2107–2108 splice/development modifications apply only to engineered masonry	Flag every masonry work package (shaft, partition, façade, landscape) on the drawing index
Seismic Design Category	Unconfirmed	2106.1 sends masonry seismic detailing to SBC 305 Chapter 7 by SDC; fireplace/chimney rebar in 2111.4 / 2113.3 is SDC **C or D** only	Lock SDC on the code datum sheet before detailing masonry or fireplaces
Ceramic-tile setting method	Unconfirmed	2103.2.3–2103.2.3.7 and Table 2103.2.3 apply only if ceramic tile is set in the listed mortars/adhesives	Freeze the tile specification (thick-bed, thin-set, epoxy, organic adhesive)
Glass unit masonry	Unconfirmed	2110 bans glass block in fire/smoke walls except listed opening-protective / atrium branches	Confirm whether lobby, stair or amenity glass block is proposed
Masonry fireplace / heater / chimney	Unconfirmed; possible in units or amenity	Sections 2111–2113 millimetre geometry apply only if those appliances are specified	Freeze fireplace/heater/chimney presence before using Sections 9–11
Adobe empirical (2109)	Not opened	Type V, **one-storey** (two-storey only with RDP), wind (V_{asd}) **49 m/s** limit — not a high-rise system	Do not apply 2109 or Table 2109.2.3.1 to this tower
Dry-stack (2114)	Not opened	Unreinforced surface-bonded CMU is not a typical high-rise wall system; Table 2114.4 is flattened OCR	Do not apply 2114 unless the SER proposes dry-stack landscape/site walls as a separate package

Decision / gap	Current project basis	Why it controls Chapter 21 application	Required project action
Special inspections	Outbound to Chapter 17	2101.3 and 2105.1 send masonry SI/tests to Chapter 17 and SBC 305	Coordinate the Chapter 17 SI schedule with the masonry specification
Sprinkler / mixed use / storeys	Unconfirmed	Chapter 21 has no occupancy, sprinkler or storey branch for engineered masonry	Do not wait on NFPA 13 vs 13R to apply 2101–2108
NOC / stamped structural	Unconfirmed	Masonry design, SCD acceptance and fireplace/chimney fire safety cannot be concluded from this extract	Engage the structural engineer of record before design freeze

4. Scope, companions, veneer and inspections

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2101.1	Chapter masonry scope	This chapter governs the materials, design, construction and quality of masonry	Masonry in the building	None stated	Direct	Treat Chapter 21 as the charging chapter; send member design to the SBC 305 package	Verified
2101.2	SBC 305 companion	Masonry shall comply with **SBC 305** as well as applicable requirements of this chapter	All masonry	None stated	External verification	Keep a live SBC 305 specification; do not import 305 thicknesses, (f'_m) or grout-space tables here	Verified
2101.2.1	Masonry veneer companion	Masonry veneer shall comply with the provisions of **Chapter 14**	Anchored or adhered masonry veneer	None stated	External verification	Detail veneer from Chapter 14; do not take veneer ties, cavities or adhesive thicknesses from this chapter	Verified
2101.3	Masonry special inspection	Special inspection of masonry shall be as defined in **Chapter 17**, or an itemized testing and inspection program that meets or exceeds Chapter 17	Masonry construction	Itemized program permitted if it meets or exceeds Chapter 17	External verification	Attach the Chapter 17 SI schedule to the masonry specification; do not invent frequencies here	Verified

5. Materials and ceramic-tile setting

Do not adopt 2103 commentary millimetres or strengths (unit densities, Grade SW **31 MPa**, thin brick **45 mm**, **75 mm** facing, thick-bed **19–32 mm**). Those numbers are commentary.

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2103.1	Masonry unit standards	Concrete, clay/shale, stone, glass-unit and AAC masonry units shall comply with **Article 2.3 of SBC 305**. Architectural cast stone: **ASTM C1364** and **TMS 504**. Adhered manufactured stone veneer units: **ASTM C1670**	Masonry units specified	Structural clay tile used only as nonstructural fireproofing or wall furring need not meet compressive-strength specifications; fire-resistance rating from **ASTM E119** or **UL 263** and **Table 705.5**	Direct	Name the 2103.1 standards on the masonry specification; send fireproofing ratings to Chapter 7 / Table 705.5	Verified
2103.1.1	Second-hand masonry units	Second-hand units shall not be reused unless they conform to new-unit requirements, are whole and sound, and have old mortar cleaned off	Salvaged or reused masonry units	None stated	Conditional	Prohibit salvaged units on the specification unless tested to new-unit criteria and cleaned of old mortar	Verified
2103.2.1	Masonry mortar	Mortar for masonry construction shall conform to **Articles 2.1 and 2.6 A of SBC 305**	Masonry mortar	None stated	External verification	Specify mortar types from SBC 305; do not import ASTM C270 proportions from commentary	Verified
2103.2.2	Surface-bonding mortar	Surface-bonding mortar shall comply with **ASTM C887**. Surface bonding of concrete masonry units shall comply with **ASTM C946**	Surface-bonded CMU	None stated	Conditional	If surface-bonded walls are used, specify C887 mortar and C946 construction; typical tower grouted CMU does not use this path	Verified
2103.2.3	Ceramic-tile Portland cement mortar	Portland cement mortars for ceramic wall and floor tile shall comply with **ANSI A108.1A** and **ANSI A108.1B** and the compositions in **Table 2103.2.3**	Ceramic tile set in Portland cement mortar	Other 2103.2.3.1–2103.2.3.7 setting methods	Conditional	Name A108.1A/B on the tile specification; do not reconstruct Table 2103.2.3 mix parts from the flattened extract	Verify source
2103.2.3.1–2103.2.3.5	Listed ceramic-tile mortars	Dry-set **ANSI A118.1** / install **A108.5**; latex-modified **A118.4** / **A108.5**; epoxy **A118.3** / **A108.6**; furan **A118.5** / **A108.8**; modified epoxy-emulsion **A118.8** / **A108.9**	Ceramic tile using that setting system	Each standard pair applies only to its mortar type	Conditional	Match the tile mortar to the listed ANSI pair; do not import A118 shear values except 2103.2.3.6	Verified
2103.2.3.6	Organic-adhesive tile bond	Water-resistant organic adhesives shall comply with **ANSI A136.1**. Shear bond strength after water immersion **not less than 275 kPa** (Type I) and **not less than 140 kPa** (Type II). Install per **ANSI A108.4**	Ceramic tile set in organic adhesive	Type I vs Type II per A136.1	Conditional	Put **275 kPa** / **140 kPa** on the interior tile adhesive specification; do not use organic adhesive as an exterior setting bed from this chapter	Verified
2103.2.3.7	Portland cement tile grout	Portland cement grouts for ceramic tile shall comply with **ANSI A118.6** and be installed in accordance with **ANSI A108.10**	Ceramic tile grouted with Portland cement grout	None stated	Conditional	Specify A118.6 grout with A108.10 installation on wet-area tile drawings	Verified
2103.2.4	Adhered-veneer mortar	Mortar for adhered masonry veneer shall conform to **ASTM C270** Type **N or S**, or **ANSI A118.4** latex-modified Portland cement mortar	Adhered masonry veneer	Veneer construction remains Chapter 14	Conditional	Coordinate Type N/S or A118.4 with the Chapter 14 adhered-veneer detail	Verified
2103.3	Masonry grout	Grout shall comply with **Article 2.2 of SBC 305**	Grouted masonry	None stated	External verification	Take grout type and pour-height limits from SBC 305; do not import commentary pea-gravel **10 mm**	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2103.4	Unidentified reinforcement tests	Metal reinforcement and accessories shall conform to **Article 2.4 of SBC 305**. Where unidentified reinforcement is approved, **not less than three** tension and **three** bending tests on representative specimens from each shipment and grade	Reinforcement and accessories; extra tests only if unidentified steel is approved	Identified mill-certified bars do not trigger the six-test rule	Conditional	Ban unidentified rebar unless the SER schedules the **three** tension and **three** bending tests per shipment and grade	Verified

6. Construction and quality assurance

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2104.1	Masonry construction standards	Masonry construction shall comply with 2104.1.1–2104.1.2 and with **SBC 305** or **TMS 604**	Masonry construction; TMS 604 for architectural cast stone	None stated	External verification	Specify SBC 305 (or TMS 604 for cast stone) tolerances; do not import commentary **4°C** cold-weather limits	Verified
2104.1.1	Masonry support on wood	Masonry shall **not** be supported on wood girders or other wood construction except as permitted in **Section 2304.13**	Masonry bearing on wood	Only the 2304.13 permission	Direct	Keep masonry off wood structure; if a wood support is proposed, design it under 2304.13 and do not import that section's millimetres here	Verified
2104.1.2	Projecting cornice kern	Unless structural support and anchorage resist overturning, the center of gravity of projecting masonry or molded cornices shall lie within the **middle one-third** of the supporting wall. Terra cotta and metal cornices shall have a structural frame of approved noncombustible material, anchored in an approved manner	Projecting masonry or molded cornices	Engineered support/anchorage may replace the kern rule	Conditional	Keep unanchored cornice CG inside the **middle one-third**; provide a noncombustible frame for terra cotta or metal cornices	Verified
2105.1	Masonry quality assurance	A quality assurance program shall ensure constructed masonry matches the approved documents, and shall comply with the inspection and testing requirements of **Chapter 17** and **SBC 305**	All masonry construction	None stated	External verification	Put the Chapter 17 / SBC 305 QA programme on the masonry specification; do not invent prism or mortar test frequencies here	Verified

7. Seismic, allowable-stress and strength-design modifications

Apply 2107–2108 only where masonry is designed as a structural system. Typical nonstructural partitions still follow 2101–2105 and SBC 305 construction rules, but not these splice equations unless the SER uses ASD/strength design.

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2106.1	Masonry seismic design	Masonry structures and components shall comply with **Chapter 7 of SBC 305** depending on the structure's seismic design category	Masonry structures and components	None stated in this chapter	External verification	Take SDC-based masonry seismic detailing from SBC 305 Chapter 7; do not invent confinement or wall-type rules here	Verified
2107.1	ASD masonry path	ASD masonry shall comply with 2106 and **Chapters 1 through 8 of SBC 305**, except as modified by 2107.2–2107.3	Allowable stress design of masonry	None stated	Conditional	If ASD is used, design under SBC 305 Ch. 1–8 plus the 2107 lap/splice modifications	Verified
2107.2–2107.2.1	ASD lap-splice length	Alternative to SBC 305 6.1.6.1.1: (*ld = 0.29 db fs) but **not less than 300 mm**, and in no case less than **40** bar diameters. Where design tensile stress **> 80 percent** of (Fs)*, increase lap **not less than 50 percent** of the minimum required length, but need not exceed **72 db**. Equivalent 50-percent stress-transfer methods permitted. Epoxy-coated bars: increase lap **50 percent**	ASD masonry reinforcement spliced by lapping	2107.3: bars larger than **Dia 32 mm** shall not be lap spliced	Conditional	Dimension ASD laps as the greatest of Eq. 21-1, **300 mm** and **40 db**, then apply the **80 percent** / epoxy **50 percent** increases with the **72 db** cap	Verified
2107.3	Large-bar splice type	Modify SBC 305 8.1.6.7: lap, welded or mechanical splices permitted. Welding shall conform to **AWS D1.4**. Welded splices shall be **ASTM A706**. Reinforcement larger than **Dia 32 mm** shall be spliced using mechanical connections in accordance with 8.1.6.7.3	ASD masonry splices	Mechanical only above **Dia 32 mm**; welded only on A706	Conditional	Ban lap splices above **Dia 32 mm**; specify A706 if welding, otherwise mechanical couplers	Verified
2108.1	Strength-design masonry path	Strength design of masonry shall comply with 2106 and **Chapters 1 through 7 and Chapter 9 of SBC 305**, except as modified by 2108.2–2108.3	Strength design of masonry	**AAC masonry** shall comply with SBC 305 Chapters 1–7 and **Chapter 11**	Conditional	If strength design is used, follow SBC 305 plus 2108 development/splice modifications; send AAC to Chapter 11 of SBC 305	Verified
2108.2	Strength-design development length	Modify SBC 305 6.1.5.1.1: required development length by Equation (6-1) but **not less than 300 mm** and need not be greater than **72 db**	Strength-design masonry reinforcement development	Equation (6-1) lives in SBC 305; do not reconstruct it here	Conditional	Cap and floor development length at **300 mm** min / **72 db** max on strength-design masonry drawings	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2108.3	Strength-design welded and mechanical splices	Welded splice: bars butted and welded to develop **not less than 125 percent** of (f_y); **ASTM A706** only; **not permitted** in plastic hinge zones of intermediate or special reinforced walls. Mechanical splices: Type 1 or 2 per **SBC 304 Section 18.2.7.1**; Type 1 **not** in plastic hinge zones or beam-column joints of intermediate or special reinforced masonry shear walls; Type 2 permitted anywhere in a member	Strength-design masonry splices	Type 1 banned in the listed hinge/joint locations	Conditional	Specify A706 welded splices at **125 percent** (f_y) outside plastic hinges; use Type 2 couplers in hinge zones. Do not import SBC 304 coupler test values	Verified

8. Glass unit masonry

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2110.1	Glass-block companion	Glass unit masonry shall comply with **Chapter 13 of SBC 305** and this section	Glass unit masonry	None stated	External verification	If glass block is used, design panels under SBC 305 Chapter 13 plus 2110.1.1; do not import 305 panel sizes here	Verified
2110.1.1	Glass-block fire-wall ban	Solid or hollow approved glass block shall **not** be used in fire walls, party walls, fire barriers, fire partitions or smoke barriers, or for load-bearing construction. Erect with mortar and reinforcement in metal channel-type frames, structural frames, masonry or concrete recesses, embedded panel anchors, or other approved joint materials. Wood strip framing shall **not** be used in walls required to have a fire-resistance rating	Glass block walls or panels	(1) Assemblies with fire-protection rating **not less than 3/4 hour** as opening protectives per Section 716 in fire barriers, fire partitions and smoke barriers rated **1 hour or less** that do **not** enclose exit stairways/ramps or exit passageways. (2) Glass-block assemblies as permitted in **Section 404.6, Exception 2**	Conditional	Keep glass block out of fire/smoke walls and load-bearing lines unless a listed **3/4-hour** opening protective or atrium 404.6 Exception 2 path is documented	Verified

9. Masonry fireplaces

Apply this section only if a masonry (or concrete) fireplace is specified. Factory-built fireplaces listed to UL 127 follow their listing where 2111.12 Exception 1 applies.

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2111.1–2111.2	Fireplace drawings	Masonry fireplaces of concrete or masonry shall comply with this section. Construction documents shall describe location, size, construction, material thickness/characteristics, and clearances from walls, partitions and ceilings	Masonry or concrete fireplace	None stated	Conditional	Put fireplace location, firebox size, wall thickness and combustible clearances on the architectural/structural set	Verified
2111.3	Fireplace footing	Footings of concrete or solid masonry **not less than 300 mm** thick, extending **not less than 150 mm** beyond the face of the fireplace or foundation wall on all sides, founded on undisturbed earth or engineered fill. In areas not subjected to freezing, footings **not less than 300 mm** below finished grade	Masonry fireplace and its chimney	Deeper/heavier footings may be required for large units; this chapter does not reduce the minima	Conditional	Show a **300 mm** footing, **150 mm** projection and **300 mm** cover below grade on the fireplace foundation detail	Verified
2111.3.1	Ash-dump cleanout	Cleanout openings in foundation walls below fireboxes, where provided, shall have ferrous metal or masonry doors and frames that remain tightly closed except when in use, accessible, and located so ash removal does not hazard combustibles	Ash dump provided	Cleanouts are "where provided," not mandatory	Conditional	If an ash dump is used, specify a tight ferrous or masonry door clear of combustibles	Verified
2111.4–2111.4.2	Fireplace seismic reinforcement	SDC **A or B:** seismic reinforcement not required. SDC **C or D:** chimneys up to **1000 mm** wide get **four Dia 12** continuous vertical bars grouted per SBC 305 Article 202; wider chimneys get **two** additional **Dia 12** bars per extra **1000 mm** or fraction. Vertical bars enclosed in **6 mm** ties (or equivalent) at **not more than 450 mm** on centre in concrete, or in bed joints at least every **450 mm** of masonry height, with **two** ties at each bend	Masonry fireplace in SDC C or D	Not required in SDC A or B	Conditional	Once SDC is C or D, schedule **four Dia 12** plus **6 mm** ties at **450 mm**; add two bars per extra **1000 mm** width	Verified
2111.5	Fireplace seismic anchorage	Anchor at each floor, ceiling or roof line **more than 1800 mm** above grade with **two 5 mm by 25 mm** straps embedded **not less than 300 mm** into the chimney, hooked around the outer bars and extending **150 mm** beyond the bend, each fastened to **not fewer than four** floor joists with **two 12.5 mm** bolts	Masonry fireplace/foundation seismic anchorage	Not required in SDC **A or B**, or where the fireplace is constructed completely within the exterior walls	Conditional	For exterior chimneys in SDC C or D above **1800 mm**, detail the two straps, **300 mm** embed and **12.5 mm** bolts to floor framing	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2111.6	Firebox wall thickness	Fireboxes of solid masonry, grouted hollow units, stone or concrete. With firebrick lining **not less than 50 mm**: back and sidewalls **200 mm** solid masonry including the lining; firebrick joints **not greater than 6 mm**. Without lining: back and sidewalls **250 mm** solid masonry. Firebrick **ASTM C27** or **C1261** in medium-duty refractory mortar **ASTM C199**	Masonry firebox	Lined vs unlined thickness branch	Conditional	Dimension lined fireboxes at **200 mm** with **50 mm** firebrick and **6 mm** max joints, or unlined at **250 mm**	Verified
2111.6.1	Steel fireplace unit	Steel units permitted with solid masonry per listing or this section. Steel firebox lining **not less than 6 mm**, with an air-circulating chamber ducted indoors. Encase to a total back/side thickness **not less than 200 mm**, of which **not less than 100 mm** is solid masonry or concrete. Circulating ducts of metal or masonry	Steel fireplace unit in a masonry fireplace	Listing may replace these thicknesses	Conditional	If a steel insert is used, keep **6 mm** lining and **200 mm** / **100 mm** masonry encasement, or install strictly to the UL listing	Verified
2111.7	Firebox and throat size	Firebox minimum depth **500 mm**. Throat **not less than 200 mm** above the fireplace opening and **not less than 100 mm** in depth. Passageway above the firebox (throat, damper, smoke chamber) **not less than** the flue cross-sectional area	Concrete or masonry fireplace firebox	Rumford: depth **not less than 300 mm** and at least **one-third** of the opening width; throat **not less than 300 mm** above the lintel and at least **1/20** the opening area	Conditional	Set conventional fireboxes at **500 mm** depth with throat **200 mm** up and **100 mm** deep; use the Rumford **300 mm** / **1/3** / **1/20** branch only for that style	Verified
2111.8–2111.8.1	Fireplace lintel and damper	Masonry over the opening supported by a noncombustible lintel bearing **not less than 100 mm** on each end. Throat or damper **not less than 200 mm** above the top of the opening. Ferrous metal damper operable from the room; controls may be in the fireplace	Masonry fireplace opening	None stated	Conditional	Show a noncombustible lintel with **100 mm** bearing and a ferrous damper **200 mm** above the opening	Verified
2111.9–2111.9.1	Smoke chamber	Front, back and sidewalls **200 mm** solid masonry, inside parged with ASTM C199 mortar. With **50 mm** firebrick or **16 mm** vitrified clay (ASTM C315) lining: total **150 mm** including lining. Inside height from throat to flue **not greater than** the inside width of the opening. Prefabricated or sloped surfaces **not more than 45 degrees** from vertical; corbeled masonry **not more than 30 degrees** from vertical	Masonry smoke chamber	Lined **150 mm** vs unlined **200 mm**	Conditional	Limit smoke-chamber height to the opening width and slope to **45°** / **30°**; keep walls at **200 mm** or **150 mm** with lining	Verified
2111.10–2111.11	Hearth and hearth extension	Hearths and extensions of concrete or masonry on noncombustible supports, reinforced for self-weight and imposed loads; no combustible material against the underside. Hearth **not less than 100 mm** thick. Extension **not less than 50 mm** thick (**10 mm** brick/concrete/stone/tile permitted where the firebox opening is raised **not less than 200 mm**). Extension **not less than 400 mm** in front and **200 mm** beyond each side; where the opening is **0.6 m²** or larger, **500 mm** front and **300 mm** each side	Masonry fireplace hearth	Raised-opening **10 mm** thickness branch; larger-opening projection branch	Conditional	Draw a **100 mm** inner hearth and size the extension from the **0.6 m²** split (**400/200** vs **500/300** mm)	Verified
2111.12	Fireplace combustible clearance	Interior or in-wall fireplaces: **not less than 50 mm** to combustibles from front faces and sides, **not less than 100 mm** from back faces. Do not fill the airspace except fire blocking per 2111.13	Masonry fireplace adjacent to combustibles	(1) UL 127 listed-and-labeled for contact, installed to manufacturer. (2) Fireplace in masonry/concrete walls: combustibles **not less than 300 mm** from the inside of the nearest firebox lining. (3) Exposed trim/sheathing may abut sidewalls and hearth extension per Figure 2111.12 if **not less than 300 mm** from the lining. (4) Combustible mantel/trim on the front: **not within 150 mm** of the opening; material directly above and within **300 mm** of the opening shall not project more than **3 mm** for each **25 mm** from the opening; side projections **more than 40 mm** need extra clearance equal to the projection	Conditional	Hold **50 mm** / **100 mm** airspace unless a listed or **300 mm**-from-lining exception is detailed; check mantel projection with the **3 mm** / **25 mm** rule	Verified
2111.13	Fireplace fire blocking	Spaces between fireplaces and floors/ceilings through which they pass shall be fire blocked with noncombustible material securely fastened. Wood-joist spaces blocked to a depth of **25 mm**, only on metal or metal lath across the space	Fireplace passing through floor or ceiling	None stated	Conditional	Fire-block each floor/ceiling penetration; keep the wood-joist block **25 mm** deep on metal lath	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2111.14–2111.14.6	Exterior combustion air	Factory-built or masonry fireplaces in this section shall have an exterior air supply unless the room is mechanically ventilated to neutral or positive pressure. Intake from outdoors, not from garage/attic/basement/crawl space, not higher than the firebox, with corrosion-resistant **6 mm** mesh. Unlisted ducts: **25 mm** clearance to combustibles within **1500 mm** of the outlet. Passageway **not less than 3870 mm²** and **not more than 0.035 m²** except listed/tested systems. Outlet in the back or sides of the firebox or within **600 mm** of the opening on or near the floor, closable, and designed so burning material cannot drop into concealed combustibles	Fireplace requiring exterior combustion air	Mechanical ventilation to neutral/positive pressure waives the supply. Listed ducts follow listing. Listed/tested fireplaces may use manufacturer passageway size	Conditional	Provide an outdoor intake with **6 mm** mesh, **3870 mm²–0.035 m²** passageway and a closable outlet within **600 mm** of the opening, unless the HVAC pressure exception applies	Verified

10. Masonry heaters

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2112.2	Masonry heater standard	Install per this section and either **ASTM E 1602**, or listed and labeled **UL 1482** or **EN 15250** to the manufacturer's instructions	Masonry heater	Two compliance paths	Conditional	If a masonry heater is specified, pick ASTM E1602 or a UL 1482 / EN 15250 listing before detailing	Verified
2112.3	Heater firebox floor	Firebox floor **minimum thickness of 100 mm** of noncombustible material, supported on a noncombustible footing and foundation in accordance with 2113.3	Masonry heater	Foundation geometry from 2113.3	Conditional	Show a **100 mm** noncombustible firebox floor on the 2113.3 footing	Verified
2112.4	Heater seismic reinforcing	SDC **D, E or F:** anchor the heater to the masonry foundation per Section F3. Seismic reinforcing is **not** required within the heater body where height **≤ 3.5 times** body width and the chimney is not supported by the heater body. Where the chimney shares a wall with the heater facing, reinforce the chimney per 2113	Masonry heater in SDC D, E or F	Squat-body exemption; chimney-share triggers 2113	Conditional	In SDC D–F, check the **3.5** aspect ratio and chimney support before omitting body reinforcement	Verified
2112.5	Heater combustible clearance	Combustibles **not within 900 mm**, or the NFPA 211 Section 12.6 allowed reduction, fully vented for free airflow around all heater surfaces	Masonry heater	(1) Wall **not less than 200 mm** solid masonry and heat-exchange channels **not less than 125 mm** solid masonry: combustibles **not within 100 mm**; gas-tight capping slab **not less than 200 mm** from a combustible ceiling. (2) UL 1482 or EN 15250 listed, installed to manufacturer	Conditional	Hold **900 mm** unless the **200/125 mm** masonry or listing reduction is documented, including **200 mm** to a combustible ceiling	Verified

11. Masonry chimneys

Apply with Section 9 when a masonry fireplace is specified, or independently for a masonry chimney serving another appliance. Medium- and high-heat appliance chimneys are not typical in Group R-2 dwelling units; keep those rows Conditional on a central plant or similar.

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2113.1–2113.2	Chimney footing	Masonry chimneys of solid masonry, grouted hollow units, stone or concrete. Footings of concrete or solid masonry **not less than 300 mm** thick, extending **not less than 150 mm** beyond the foundation or support wall on all sides, on undisturbed earth or engineered fill; **not less than 300 mm** below finished grade where not subjected to freezing	Masonry chimney	None stated	Conditional	Match the fireplace footing: **300 mm** thick, **150 mm** projection, **300 mm** below grade	Verified
2113.3–2113.3.2	Chimney seismic reinforcement	SDC **A or B:** not required. SDC **C or D:** chimneys up to **1000 mm** wide get **four Dia 12** continuous vertical bars grouted per SBC 305 Article 2.2, with grout kept from bonding to the flue liner; wider chimneys get **two** additional **Dia 12** per extra **1000 mm**. **6 mm** ties at **not more than 450 mm** in concrete or every **450 mm** of masonry height; **two** ties at each bend	Masonry chimney in SDC C or D	Not required in SDC A or B	Conditional	Same vertical/tie schedule as 2111.4; keep the liner free to move thermally	Verified
2113.4	Chimney seismic anchorage	Anchor at each floor/ceiling/roof line **more than 1800 mm** above grade with **two 5 mm by 25 mm** straps embedded **not less than 300 mm**, hooked around outer bars, extending **150 mm** beyond the bend, each to **not fewer than four** floor joists with **two 12.5 mm** bolts	Masonry chimney seismic anchorage	Not required in SDC **A or B**, or where the masonry fireplace is constructed completely within the exterior walls	Conditional	Detail exterior-chimney straps above **1800 mm** in SDC C or D	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2113.5–2113.6	Chimney corbel and floor crossing	Do not corbel more than **half** the chimney wall thickness from a wall or foundation, nor from a wall/foundation **less than 300 mm** thick unless the chimney projects equally on each side, except on the **second story** of a two-story dwelling, exterior corbeling may equal the wall thickness. Single-course projection **not more than one-half** the unit height or **one-third** the unit bed depth, whichever is less. Do not change chimney wall or flue size/shape within **150 mm** above or below a floor, ceiling or roof crossing	Masonry chimney corbel or level change	Equal projection both sides waives the **300 mm** wall minimum; two-story dwelling exterior exception	Conditional	Limit corbels to **½** wall thickness and keep size/shape changes **150 mm** clear of combustible floor/roof assemblies	Verified
2113.8–2113.9.1	Chimney termination and cap	Chimneys shall not support loads other than self-weight unless designed for the additional load. Extend **not less than 600 mm** higher than any portion of the building within **3000 mm**, and **not less than 900 mm** above the highest point where the chimney passes through the roof. Provide a concrete, metal or stone cap sloped to shed water, with drip edge and caulked bond break around flue liners per **ASTM C 1283**	Masonry chimney termination	Medium- and high-heat appliances have taller 2113.11.2.4 / 2113.11.3.3 limits	Conditional	Set residential chimneys at **900 mm** above the roof and **600 mm** above anything within **3000 mm**; add a sloped cap per C1283	Verified
2113.9.2–2113.9.3	Spark arrestor and rain cap	If provided: arrestor net free area **not less than four times** the flue outlet; screen equivalent to **19-gage** galvanized or **24-gage** stainless; openings that block **12.5 mm** spheres but pass **10 mm** spheres; accessible/removable. Rain cap net free area under the cap **not less than four times** the flue outlet	Spark arrestor or rain cap installed	Neither device is mandated by this section	Conditional	If a cap or arrestor is used, keep **4×** outlet area and the **10–12.5 mm** mesh	Verified
2113.10–2113.10.1	Chimney wall thickness	Walls of concrete, solid masonry or grouted hollow units with **not less than 100 mm** nominal thickness. Masonry used as veneer on a framed chimney: through flashing and weeps as required by **Chapter 14**	Residential-type / fireplace chimney walls	Medium-heat **200 mm** / stone **300 mm**; high-heat double **200 mm** walls	Conditional	Keep ordinary chimney wythes at **100 mm** nominal; send veneer flashing to Chapter 14	Verified
2113.11–2113.11.1.5	Residential flue lining	Masonry chimneys shall be lined, appropriate to the appliance listing. Residential-type liners: **ASTM C 315** clay, listed **UL 1777**, factory-built chimney units listed for installation in masonry chimneys, or other approved materials resisting flue gases and condensate to **982°C**. Gas appliances: **SBC 1201**. Pellet: 2113.11.1 or listed pellet vents. Oil L-vent: 2113.11.1 or **UL 641**. Relined flues not complying with 2113.11.1 shall carry a permanent use/fuel label	Chimney serving a residential-type appliance	Appliance-specific lining paths	Conditional	Specify a C315, UL 1777 or listed liner; send gas appliances to SBC 1201; label any non-2113.11.1 reline	Verified
2113.11.2–2113.11.2.5	Medium-heat chimney	Solid masonry or concrete walls **not less than 200 mm**, or stone **not less than 300 mm**. Medium-duty refractory brick lining **not less than 110 mm** thick laid on the **110 mm**, starting **600 mm** or more below the lowest connector; chimneys terminating **7500 mm or less** above the connector lined to the top. Multiple passageways separated by **not less than 100 mm** concrete or solid masonry. Termination **not less than 3000 mm** higher than any building portion within **7500 mm**. Clearance to combustibles **not less than 100 mm**	Chimney serving a medium-heat appliance	Not typical for dwelling-unit fireplaces	Conditional	If a medium-heat appliance is proposed, use **200/300 mm** walls, **110 mm** lining, **3000/7500 mm** termination and **100 mm** clearance	Verified
2113.11.3–2113.11.3.4	High-heat chimney	Double walls of solid masonry or concrete, each **not less than 200 mm** with **not less than 50 mm** airspace. High-duty refractory brick lining **not less than 110 mm** from the base to the top. Termination **not less than 6100 mm** higher than any building portion within **15200 mm**. Approved clearance from buildings for overheating, inspection and burns	Chimney serving a high-heat appliance	Not typical for Group R-2 dwelling units	Conditional	Do not use a high-heat chimney on this tower unless a plant appliance is confirmed; then apply double **200 mm** walls, **50 mm** cavity and **6100/15200 mm** termination	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2113.12–2113.14	Clay liner install and multiple flues	Clay liners per **ASTM C 1283**, from **not less than 203 mm** below the lowest inlet (or from the top of the smoke chamber for fireplaces) to above the enclosing walls, slope **not greater than 30 degrees** from vertical, in medium-duty non-water-soluble refractory mortar **ASTM C 199**, with an airspace or insulation **not exceeding** the flue-liner thickness. Two or more flues: masonry wythes **not less than 100 mm** thick bonded into the chimney. Exception: two flues serving only one appliance may adjoin with lining only, joints staggered **not less than 100 mm**	Clay flue lining; multiple flues in one chimney	One-appliance adjoining-flue exception	Conditional	Start clay liners **203 mm** below the inlet, keep slope **≤ 30°**, and separate extra flues with **100 mm** wythes	Verified
2113.15–2113.16.2	Chimney flue area	Appliance flues **not smaller** than the connector; multiple appliances: largest connector plus **50 percent** of additional connector areas (except oil per **NFPA 31** or gas per **SBC 1201**). Fireplace flues: 2113.16.1 or 2113.16.2. Round **not less than 1/12** of the fireplace opening; square **1/10**; rectangular aspect **less than 2 to 1 1/10**; rectangular **2 to 1 or more 1/8**. Alternative: Figure 2113.16 with Table 2113.16(1) and (2) net areas, or manufacturer/field measurement	Chimney flue sizing	Oil NFPA 31 and gas SBC 1201 size exceptions. Tables 2113.16(1)–(2) are flattened OCR	Conditional	Size fireplace flues from the **1/12–1/8** ratios in 2113.16.1 until the published Figure/tables are verified; do not adopt concatenated table millimetres	Verify source
2113.17–2113.18	Chimney inlet and cleanout	Inlets from the side with a fireclay, rigid refractory or metal thimble that keeps the connector from pulling out or passing beyond the liner. Cleanout within **150 mm** of the base of each flue; upper edge **not less than 150 mm** below the lowest inlet; opening height **not less than 150 mm**; noncombustible cover	Masonry chimney inlets and cleanouts	Fireplace flues where cleaning is possible through the fireplace opening	Conditional	Place a side thimble and a **150 mm** cleanout **150 mm** below the lowest inlet, unless the fireplace opening is the cleanout	Verified
2113.19	Chimney combustible clearance	Interior or in-wall chimneys: **not less than 50 mm** airspace to combustibles. Entirely outside (including soffit/cornice): **not less than 25 mm**. Do not fill except fire blocking per 2113.20	Masonry chimney adjacent to combustibles	(1) UL 1777 liner listed for contact, installed to manufacturer. (2) Chimney in masonry/concrete walls: combustibles **not less than 300 mm** from the inside of the nearest flue lining. (3) Exposed trim/sheathing may abut sidewalls per Figure 2113.19 if **not less than 300 mm** from the lining; trim shall not overlap chimney corners by more than **25 mm**	Conditional	Hold **50 mm** inside / **25 mm** outside unless a listed or **300 mm**-from-liner exception is detailed	Verified
2113.20	Chimney fire blocking	Spaces between chimneys and floors/ceilings shall be fire blocked with noncombustible material securely fastened. Wood-joist spaces self-supporting or on metal/metal lath	Chimney passing through floor or ceiling	None stated	Conditional	Fire-block every chimney floor/ceiling penetration on metal lath at wood framing	Verified

12. Project-use controls

1. Use **Verified** rows for initial scoping after the row trigger and branch are confirmed (masonry actually present; fireplace/heater/chimney actually specified).
2. Treat every **Verify source** row as a design hold point; no affected value is to be placed in issued-for-approval drawings without a published-source check.
3. Do not apply 2109 adobe (Type V, one-/two-storey) or 2114 dry-stack to this R-2 high-rise.
4. Do not import SBC 305 wall thicknesses, (f'_m), grout-space or standard lap tables; Chapter 14 veneer millimetres; Chapter 7 / Table 705.5 ratings; Chapter 16 loads; Chapter 17 SI frequencies; ANSI A108/A118 values other than the **275 kPa** / **140 kPa** organic-adhesive stresses; or 2103 commentary densities and Grade SW strengths.
5. Keep the published tokens **Dia 32 mm**, **3/4 hour**, **3870 mm²**, **0.035 m²**, **1/12**, **1/10**, **1/8**, **982°C** and **2 1/2 percent** intact (the last is adobe-only and is not used in project-use tables).
6. Record SDC, masonry role, ceramic-tile method, glass block, and fireplace/heater/chimney decisions in the project Golden Thread; this matrix is not evidence of SCD NOC or stamped compliance.

13. Coverage summary

Internal inventory of the attached Chapter 21 extract (numbered code, exceptions, tables, footnotes; commentary excluded). Row-level records are not published.

- **Inventory scope:** numbered code, exceptions, tables, footnotes (commentary excluded)
- **Total independently checkable numeric records:** 241
- **Verified:** 202
- **Verify source:** 39
- **Appended tables:** 5 (all concatenated OCR)

Counts by top-level section

Top-level section	Records
2101	0
2102	0
2103	8
2104	1
2105	0
2106	0
2107	8
2108	3
2109	56
2110	2
2111	60
2112	7
2113	91
2114	5

2109 adobe (**50** Verified + **6** Table 2109.2.3.1 Verify source) and 2114 dry-stack (**1** Verified + **4** Table 2114.4 Verify source) are included in these counts and omitted from Sections 4–11.

Coverage cross-check against `SBC 201 Chapter 21 Masonry (2024)_CS.md` was topics-only: chapter scope; SBC 305 companion; veneer to Chapter 14; fire to Chapter 7; foundations to Chapter 18 / SBC 303; loads to Chapter 16 / SBC 301; special inspections to Chapter 17; adobe empirical limits. No CS.md value was copied into a matrix cell.

14. Unresolved-source register

Affected table / clause	Why unverified	Control note
Table 2103.2.3	Concatenated HTML; location/mortar/composition cells not independently readable	Do not adopt scratch-coat, wall, floor or ceiling mix parts. Specify ANSI A108.1A/B and verify the published table before mixing thick-bed mortar
Table 2109.2.3.1	Concatenated HTML; 12.5 mm row is dashes; remaining diameter/embedment/shear cells run together	Adobe not opened for this high-rise. Do not reconstruct bolt shears
Table 2113.16(1)	Concatenated HTML; inside diameter and area cells run together. Footnote a names ASTM C 315	Use 2113.16.1 ratios (**1/12**, **1/10**, **1/8**) until the published round-flue table is verified. Do not adopt concatenated **mm²** cells
Table 2113.16(2)	Concatenated HTML; nominal outside dimensions and areas run together	Same control as Table 2113.16(1) for square/rectangular liners
Table 2114.4	Concatenated HTML; compression/flexure/shear MPa cells run together	Dry-stack not opened. Do not adopt **0.31 / 0.21 / 0.12 / 0.07 MPa** from the flattened extract
Figure 2113.16	Figure present in the extract as a caption only; no readable plot values	Do not invent chimney-height/opening-area coordinates. Size from 2113.16.1 or a verified published figure
Figures 2111.12, 2111.12(1), 2113.19	Captions only; exception geometry is in the text	Use the numbered exception millimetres in 2111.12 and 2113.19; do not invent figure-only offsets